

# JUSTIN NGUYEN

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1-619-622-9072

Computer Science undergraduate with experience in C++, Python, and Linux and a strong grasp of algorithms, data structures, and systems programming. Built low-latency, networked applications and real-time tools. Eager to contribute to high-performance software.

## PROJECTS

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- **Language Learning ChatGPT chatbot**
  - Designed and developed a conversational AI chatbot using Python and OpenAI's ChatGPT API to generate dynamic, context-aware responses
  - Integrated speech-to-text and text-to-speech libraries for a natural user experience and real-time voice interactions
  - Built a user-friendly GUI using Tkinter, applying principles of user-centered design and event-driven programming
- **Multiplayer Card Game -Python Networking Project**
  - Developed a real-time multiplayer card game using Python and TCP socket programming to enable client-server communication
  - Configured a long-running Python socket server on AWS EC2 using Linux tools (screen, SO\_REUSEADDR) to enable reliable client connections and uptime
  - Handled asynchronous game state updates and serialization errors
- **Unix Shell**
  - Designed a shell program in C++ that followed UNIX system calls such as fork(), exec(), wait(), dup2(), and pipe() to create processes that run concurrently
  - Implemented command parsing and tokenization in order to handle user input, to support command chaining and redirection

## EMPLOYMENT

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- | Software Engineer Intern                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Sigma Connectivity | July 2022 - Sept 2022 |
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| <ul style="list-style-type: none"><li>• Worked on the development of mobile applications by using the Android Open-Source Project (AOSP), while employing best practices in software design to guarantee scalability, maintainability, and optimal performance</li><li>• Performed hands on debugging and testing procedures through flashing and building code on a physical device (Google Pixel 5a)</li><li>• Implemented version control using git, navigating and manipulating projects in a Linux environment</li></ul> |                    |                       |

## CORE SKILLS

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Python, C++, Linux, Java, SQL, Git, Android (AOSP), Bash Code, Network Programming, Embedded Programming, API Programming, AWS, Data Structures and Algorithms, OS Design Knowledge, Multithreading, Automation

## EDUCATION

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- | San Marcos, CA                                                                          | Cal State San Marcos | January 2022 – May 2025 |
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| <ul style="list-style-type: none"><li>• Bachelor of Science, Computer Science</li></ul> |                      |                         |